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## Design and modeling of a driving system for soft massage robot<sup>☆</sup>

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### ABSTRACT

In this paper, a driving system of the soft massage robot was presented, capable of switching between curved and vertical states to complete palm kneading and palm pressing. The bending and the vertical state were achieved by filling unequal-pressure and isobaric gas into 3 actuators respectively. On that basis, first, the theoretical model between  $F$  and  $L$ ,  $D$ ,  $D_1$ ,  $D_2$ ,  $h_1$ ,  $P_{jin}$  at the vertical state was built by complying with the force balance equation and the moment balance equation. Second, the actuator was simplified as an equivalent spring, and then the kinematics model at the bending state was established. Lastly, the output force model at the bending state was set in accordance with the force balance equation again. To verify the accuracy of the workspace and the output force, a set of experimental equipment was built, consisting of a pneumatic module prototype, a pneumatic control system, as well as an experimental testing device. Furthermore, the stability of the pneumatic module performance was determined by performing the repeated experiments, and the consistency between the theoretical model and the experimental results was verified.

### 1. Introduction

Over the past few years, China's aging population has surged, and the elderly are encountering a wide range of chronic diseases, thereby seriously affecting their health. Among them, back pain refers to a typical disease of senior citizens [1–3]. With the fast-paced lifestyle and the lack of necessary physical exercise, aching backs have also begun to be a common disease among young people [4]. For its long history and mature technology, traditional Chinese medicine massage has been considered an effective physical therapy for patients to relieve and treat back pain [5]. By conducting an experimental research, Jiang [6] reported that during a back massage, kneading and pressing the painful part can mitigate the patient's discomfortable symptoms.

At present, most types of massage actions are achieved by massage physicians. To address the short supply of massage physicians, massage robots progressively access into our daily life [7,8] to help people eliminate fatigue and, prevent and treat diseases. Many institutions and enterprises have conducted relevant research, such as Hu [9] proposed a massage robot based on traditional Chinese medicine massage therapy that could perform multiple massage movements. Sergey [10] designed a self-portable and intelligent massage robot by complying with a planar parallel mechanism, Inconsistent with other industrial robots and

multi-module parallel robots, the sliding of multiple vacuum massage cups on it is used to simulate the cupping massage on the back. The facial massage robot developed by Atsuo [11,12], capable of scanning the human face via the thermal infrared imager and then determining the appropriate massage action and massage strength according to the scanning results. Zeng [13] developed 5 degrees of freedom massage robot exhibiting a simple structure and a complete decoupling of motion control. The moving guide, the parallel connection actuators, and the massage hand cooperating with each other can complete four types of massage movements, patting, pushing, rolling, and rubbing.

Most of the currently developed massage robots are rigid robots composed of rigid components that are capable of generating the force required to satisfy the massage. The intensity is optimal between 8 and 30N during the back massaging [14]. The suitable output force for poor physique people (elderly and children) reaches 20N, and those for most people and the crowd who need larger massage force are 27N and 35N respectively. The massage acts as a form of human-computer interaction, whereas rigid massage robots are mostly passive and extrusive. Thus the consumers fail to fully trust the safety and effect of massage [15]. Inconsistent with rigid robots, scientists set up soft robots based on bionics [19] by studying the flexibility characteristics (octopus tentacles, snakes and elephant trunks [16–18]). Compared with rigid robots,

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soft robots are safer [20], and they are more suitable to contact with the human body directly for the massage.

However, the existing soft robot is limited by its own property, which cannot satisfy the higher output force required by the massage [21]. On the whole, the existing soft massage products are mostly concentrated in the areas requiring small output forces, such as the eyes, the neck, or the hands [22–25]. Given the mentioned analysis, this paper designed a driving system of soft massage robot capable of providing a higher output force and completing two massage movements, the palm kneading and the palm pressing. During the palm kneading, the massage head can be rotated in all directions at a fixed bending angle, and the force people require can be generated. The rest of this paper is organized as follows. In Section 2, the structure and function of the soft massage robot are described. In Section 3, the kinematics model and the force model of the driving system are analyzed. In Section 4, the workspace of the driving system is explored. In Section 5, the experiment and results are presented. Lastly, Section 6 summarizes the whole paper.

## 2. Structure and function of the robot

When the soft massage robot works, the massage head moves to the designated acupoints on the back of the human body via the rigid arm, and the effective massage conduct by regulating the air pressure. The soft massage robot designed in this paper is composed of a driving system and a massage head. An applying force platform, a fixed platform, and three actuators evenly distributed between the applying force platform and the fixed platform constitute the driving system, in which the three actuators exhibit the identical structural size and are 120° each other. The structure of the actuator (Fig. 1d) consists of the air pipeline, an upper-end cover, a lower-end cover, and an actuator body. The actuator body is a cylinder with a constant-cross section in which four driving cavities and a stiffness adjusting cavity are contained. The axis of the stiffness adjusting cavity is coincident with the axis of the actuator, the four identical driving cavities are evenly distributed around the stiffness adjusting cavity, and they are in parallel connections with each other. The stiffness adjusting cavity is structured as a cylindrical through-hole, of which the function is mainly shown as follows: filling the high-pressure gas into it can increase the load-bearing bending moment of the actuator means increasing the stiffness of the actuator, so that the dynamic adjustment of its stiffness is achieved. The driving cavity is structured as a semicircular through-hole which is the main source of output force of the actuator. Restraint-springs were wrapped around the outside of the actuator body to prevent radial deformation. Moreover, the actuator body, the upper-end cover, the lower-end cover, and the air pipeline form a series relationship. Generally speaking, series and parallel exist simultaneously in a soft robot that can have a higher degree of bionics than other soft robots.

The function of the soft massage robot is illustrated in Fig. 1. When the air pressure is not connected, the overall state of the soft massage robot is presented in Fig. 1a. When the air pressure inside the three actuators (Fig. 1b) is the same, the soft massage robot will deform in the vertical direction and complete the palm pressing. When one or more actuators (Fig. 1c) are filled with different air pressures, the soft massage robot will achieve the bending deformation and complete the palm kneading.

When one or more actuators are filled with different air pressures, the driving system will bend like a circular arc. Therefore the circular arc fitting was conducted for points on the central axis of the driving system. As shown in Fig. 2.

Where  $r$  denotes the radius of the fitting circle. Comparing the distance from the point on the central axis to the center of the circle with the radius of the fitting circle, the relative error of the bending circular arc could be determined. The error was 0.47% with a single actuator and 0.85% with two actuators, which demonstrated that when the driving system was in motion, the circular arc bending could be achieved along the side where the pressure was lower.

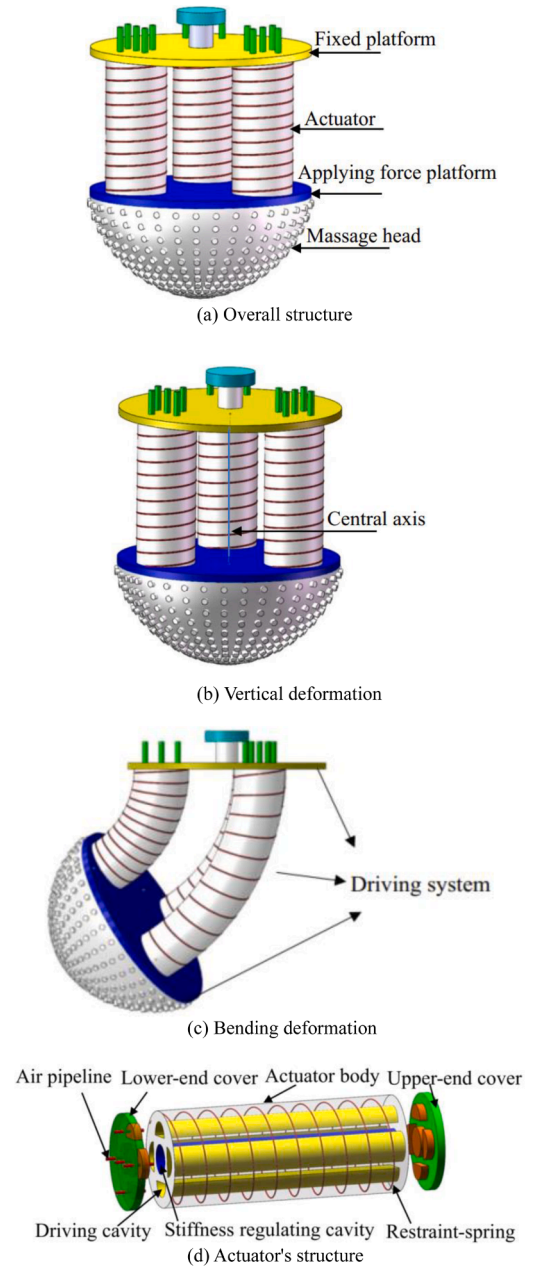


Fig. 1. Soft massage robot's structure.

## 3. Kinematics model and force model of the driving system

The driving system comprises two states of motion, the vertical deformation, and the bending deformation. The force exerted by the soft massage robot on the human body is its output force.

### 3.1. Force analysis at the vertical state

The dynamics analysis on a single actuator was conducted, in which the force state was shown in Fig. 3. Thus,

$$F_j = \sum_{i=1}^5 F_{pji} - F_{ja} - F_{jPatm} - F_{jT} \quad (j = 1, 2, 3) \quad (1)$$

Where  $F_j$  denotes the output force of the  $j$ th actuator,  $F_{pji}$  expresses the pressure on the top of the cavity after the  $i$ th chamber of  $j$ th actuator is inflated,  $F_{ja}$  represents the elastic force generated at the vertical state of

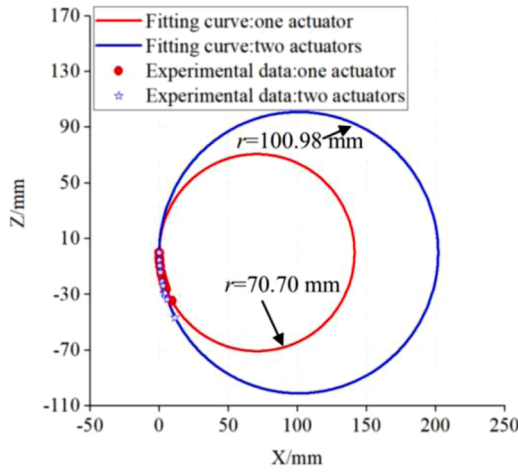


Fig. 2. Bending circular arc of the central axis.

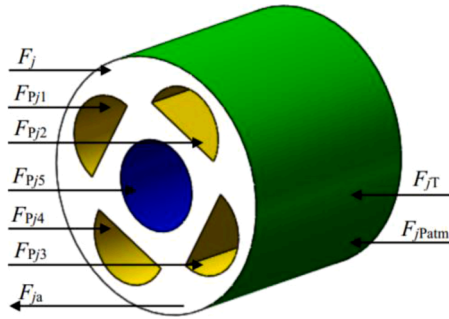


Fig. 3. Force analysis of actuator.

$j$ th actuator,  $F_{jPatm}$  is the pressure on the top of the  $j$ th actuator chamber caused by atmospheric pressure,  $F_{jT}$  denotes the impedance force produced by the restraint-spring of the  $j$ th actuator at the vertical state.

The single actuator consists of 4 driving cavities and 1 stiffness regulating cavity, all of the driving cavities will be filled with the same gas in the presence of only axial deformation and no bending deformation. This paper assumed that,  $P_{ji}=P$ , where  $P_{ji}$  denotes the air pressure charged into the  $i$ th chamber of the  $j$ th actuator. Thus,

$$\sum_{i=1}^5 F_{pji} = (S_{j1} + S_{j2} + S_{j3} + S_{j4} + S_{j5})P \quad (2)$$

Where  $S_{ji}$  denotes the pressurized area at the top of the  $i$ th chamber of the  $j$ th actuator. By analyzing the model for the deformation of the restraint-spring wrapped outside the actuator, the relationship for the diameter of the restraint-spring before and after deformation was deduced.

$$D'' = \sqrt{(D')^2 - \frac{L_0^2 - L^2}{n^2 \pi^2}} \quad (3)$$

Where  $D'$  is the diameter of the restraint-spring before the deformation of the actuator and  $D'=D$ ,  $D$  represents the diameter of the actuator.  $D''$  expresses the diameter of the restraint-spring after the deformation of the actuator,  $L$  is the initial height of the actuator.  $L_0$  denotes the post-deformation height of the actuator.  $n$  is the ring number of winding of the restraint-spring. When  $n$  reached a certain value, the radial deformation of the restraint-spring could be nearly ignored in the process of elongation through the analysis of (3). Thus, it was assumed that  $D$  does not change, and the radial expansion of each driving cavity could also be neglected. Thus in (2),

$$\begin{cases} S_{j1} = S_{j2} = S_{j3} = S_{j4} = \frac{\pi D_2^2}{8} \\ S_{j5} = \frac{\pi D_1^2}{4} \end{cases} \quad (4)$$

Where  $D_1$  is the diameter of the stiffness adjusting cavity,  $D_2$  denotes the diameter of the driving cavity. Similarly, the atmospheric pressure  $P_{atm}$  produced pressure on the top of the  $j$ th actuator.

$$F_{jPatm} = (S_{j1} + S_{j2} + S_{j3} + S_{j4} + S_{j5})P_{atm} \quad (5)$$

The elastic force generated by the  $j$ th actuator at the vertical state is,

$$F_{ja} = E\epsilon A_{js} \quad (6)$$

Where  $E$  denotes Young's modulus of silica gel,  $\epsilon$  is a strain of silica gel and  $\epsilon=(L_0-L)/L=d/L$ ,  $d$  represents the amount of elongation of the actuator after deformation,  $A_{js}$  expresses the effective cross-sectional area of the silica gel at the top of the  $j$ th actuator. When the effective cross-sectional area was calculated, the cross-section at the top of the actuator would be considered a solid circle firstly and the cross-sectional area corresponding to the driving cavity and the stiffness adjusting cavity was calculated, respectively. Finally, the area of the two was subtracted. Thus,

$$A_{js} = \frac{(D^2 - D_1^2)}{4} \pi - \frac{\pi D_2^2}{2} \quad (7)$$

By complying with Hooke's law, this paper assumed that the axial deformation of the restraint-spring and the actuator is synchronized, the deformation force conforms to linear superposition, and the effect of coupling force between the two was ignored. Subsequently, at the vertical state of the  $j$ th actuator, the impedance force of the restraint-spring is,

$$F_{jT} = \frac{G_T d_1^4 d}{8 D_3^3 n} \quad (8)$$

Where  $G_T$  is the shear modulus of elasticity of the restraint-spring,  $D_3$  denotes the pitch diameter of the restraint-spring,  $d_1$  expresses the diameter of the restraint-spring. The output force of the  $j$ th actuator could be calculated by substituting (2), (4), (5), (6), (7) and (8) into (1). Thus,

$$F_j = \left( \frac{\pi D_2^2}{2} + \frac{\pi D_1^2}{4} \right) P_{jin} - \frac{\pi E d (D^2 - D_1^2)}{4L} + \frac{\pi E d D_2^2}{2L} - \frac{G_T d_1^4 d}{8 D_3^3 n} \quad (9)$$

Where  $P_{jin}=P-P_{atm}$ ,  $P_{jin}$  represents the difference between the air pressure that input into the  $j$ th actuator and the atmospheric pressure. In the axial elongation of the  $j$ th actuator, the spring wrapped outside the actuator only as a restraint ring to limit the radial expansion without generating an impedance moment. Accordingly, the  $j$ th actuator is only subjected to the moment relative to the position of the top center of the actuator which generated by  $F_j$ ,  $P_{jin}$ , and  $F_{ja}$ , they are  $M_{Fj}$ ,  $M_{Pjin}$ , and  $M_{Fja}$  respectively. Given the moment equilibrium equation, it yields,

$$M_{Fj} = M_{Pjin} + M_{Fja} \quad (10)$$

The top cross-section of the  $j$ th actuator is illustrated in Fig. 4. Bringing in the local coordinate  $h$  when the air pressure acts in the top of the actuator and finally reaches a steady state. Subsequently,  $M_{Fja}$  and  $M_{Pjin}$  could be obtained according to Fig. 4. Where  $M_{Fja}$  was calculated in the same way as  $F_{ja}$ ,  $M_{Fj}$  could be calculated on the bottom surface of the upper-end cover of the  $j$ th actuator.

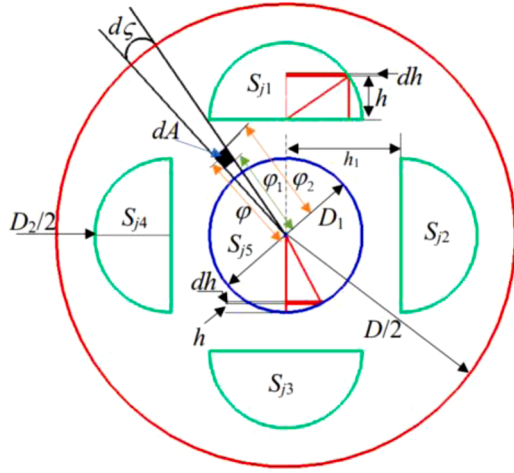


Fig. 4. Actuator cross-section.

$$\begin{cases} M_{Fja} = \int_{\zeta}^{\zeta+d\zeta} \int_{\phi_1}^{\phi_2} \frac{Ed}{L} \phi^2 d\phi d\zeta \\ M_{Fj} = \int_{\zeta}^{\zeta+d\zeta} \int_{\phi_1}^{\phi_2} F_{ja} d\phi \\ M_{Pjin} = \int_0^{\frac{D_2}{2}} 2 \cdot \frac{D_2}{2} \cdot \frac{\sqrt{D_2^2 - 4h^2}}{D_2} (h_1 + h) P_{jin} dh \end{cases} \quad (11)$$

Where  $h_1$  denotes the straight-line distance of the center of the circle between the driving cavity and the stiffness adjusting cavity. Substituting (11) into (10), and the integral calculation was conducted.

$$F_j = \frac{Ed}{\pi DL} \left( \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2} \right) + \left( \frac{D_2^3}{3} + \frac{\pi D_2^2 h_1}{2} \right) \frac{P_{jin}}{D\pi} \quad (12)$$

As impacted by the hyperelastic properties of the silica gel, the relationship between  $F_{ja}$  and  $d$  was nonlinear, and it was difficult to own the accurate expression for  $d$ . Thus, a novel expression of  $F_j$  could be gained by integrating (9) and (12).

$$\begin{aligned} F_j = & \frac{\left( \frac{(D^3 - D_1^3)}{4} \pi - \frac{\pi D_2^2}{2} + \frac{G_T d_1^4 L}{8 D_3^3 n E} \right) \cdot \left( \frac{D_2^3}{3} + \frac{\pi D_2^2 h_1}{2} \right) P_{jin}}{\left( \frac{(D^3 - D_1^3)}{4} \pi - \frac{\pi D_2^2}{2} + \frac{G_T d_1^4 L}{8 D_3^3 n E} \right) D\pi + \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2}} \\ & + \frac{\left( \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2} \right) \left( \frac{\pi D_2^2}{2} + \frac{\pi D_1^2}{4} \right) P_{jin}}{\left( \frac{(D^3 - D_1^3)}{4} \pi - \frac{\pi D_2^2}{2} + \frac{G_T d_1^4 L}{8 D_3^3 n E} \right) D\pi + \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2}} \end{aligned} \quad (13)$$

When the massage head contacted the specified acupoint, the total output force of 3 actuators was equal to the force  $F$  of the human body to the applying force platform. Thus,

$$F = \sum_{j=1}^3 F_j \quad (14)$$

### 3.1.1. Bending deformation analysis

Kelvin [26] conducted the experimental analysis on the silica gel and reported that it could be equivalent to the parallel connection of the springs and dampers when it was in motion, in which strain and stress were also the sums of the two. The damping effect of silica gel could be almost ignored when the change rate of air pressure was being reduced. Accordingly, the 3 actuators could be simplified to 3 spring model, and

the vertex of spring was located in the centroid of the cavity, as shown in Fig. 5.  $\theta$  denotes the bending angle,  $\beta$  represents the bending direction angle,  $R_0$  expresses the radius of curvature of the central axis of the driving system,  $R_1$  is the distance from the center of the applying force platform to the upper-end point of the equivalent spring, straight-line  $b-b'$  denotes the intersecting line of the fixed platform and the force-applying platform. Given the existing analysis, circular arcuate bending was done of the driving system in the presence of the non-isobaric tensile deformation.

Three equivalent springs were numbered #1, #2, and #3, their lengths were  $l_1$ ,  $l_2$  and  $l_3$  after the deformation respectively, and the lengths of the central axis after the deformation was  $l_4$ .  $l_1$ ,  $l_2$ ,  $l_3$ , and  $l_4$  as defined below.

$$l_m = L_0 + d_m (m=1, 2, 3, 4) \quad (15)$$

Where  $l_m$  is the length after the deformation of the equivalent spring,  $d_m$  denotes the amount of axial deformation of the equivalent spring. According to the geometric relationship in Fig. 5, it yields,

$$\begin{cases} l_1 = \theta(R_0 - R_1 \cos \beta) \\ l_2 = \theta \left( R_0 - R_1 \cos \left( \frac{2\pi}{3} - \beta \right) \right) \\ l_3 = \theta \left( R_0 - R_1 \cos \left( \frac{4\pi}{3} - \beta \right) \right) \\ l_4 = \theta R_0 \end{cases} \quad (16)$$

The relationship between  $\beta$ ,  $\theta$ ,  $R_0$  and  $l_1$ ,  $l_2$ ,  $l_3$ ,  $R_1$  could be drawn by solving (16).

$$\beta = \arctan \frac{\sqrt{3}(l_3 - l_2)}{l_3 + l_2 - 2l_1} \quad (17)$$

$$\theta = \frac{\sqrt{2((l_1 - l_2)^2 + (l_2 - l_3)^2 + (l_3 - l_1)^2)}}{3R_1} \quad (18)$$

$$R_0 = \frac{(l_1 + l_2 + l_3)R_1}{\sqrt{2((l_1 - l_2)^2 + (l_2 - l_3)^2 + (l_3 - l_1)^2)}} \quad (19)$$

### 3.1.2. Force analysis in bending state

Taking a single actuator for the analysis presented in Fig. 6. Where  $R_2$  is the radius of curvature of the center axis of the actuator,  $R_3$  denotes the distance from the upper-end point of the center axis to the upper-end point of the equivalent spring,  $\beta'$  expresses the bending direction angle of the actuator. The lengths after the deformation of the equivalent spring corresponding to the four driving cavities and the one stiffness adjusting cavity were  $l_1'$ ,  $l_2'$ ,  $l_3'$ ,  $l_4'$  and  $l_5'$ , respectively. A free point C was taken at the fixed end of the actuator and its position was represented with polar coordinates,  $o_1 C = \varphi$ ,  $\angle x o_1 C = \alpha$ .

The dynamics analysis of the  $j$ th actuator can be done according to

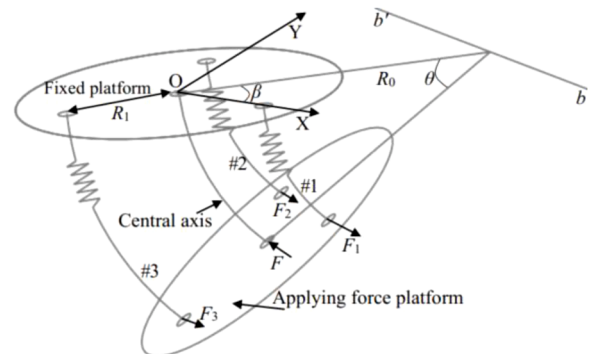


Fig. 5. Bending state analysis of the driving system.



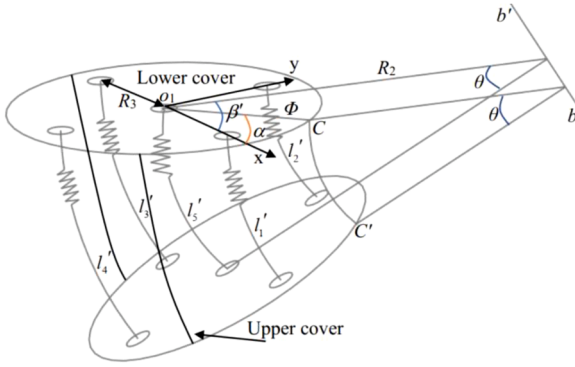


Fig. 6. Bending state analysis of the actuators.

Fig. 6.

$$F_j' = \sum_{i=1}^5 F_{Pji} - F_{ja}' - F_{j\text{atm}} - F_{jT}' \quad (20)$$

Where  $F_j'$  is the output force of the  $j$ th actuator at the bending state,  $F_{ja}'$  denotes the elastic force of the  $j$ th actuator at the bending state,  $F_{jT}'$  expresses the impedance force of the restraint-spring in the bending state of the  $j$ th actuator.

According to Fig. 4 and Fig. 6, the elastic force generated by the  $j$ th actuator at the bending state can be obtained.

$$F_{ja}' = \int_{\zeta}^{\zeta+d\zeta} \int_{\phi_1}^{\phi_2} E \left( \frac{(R_2 - \phi \cos(\beta' - \alpha))}{L} \theta - 1 \right) \phi d\phi d\zeta \quad (21)$$

In accordance with Hooke's law, the impedance force of the restraint-spring of the  $j$ th actuator at the bending state could be determined.

$$F_{jT}' = \frac{G_T d_1^4 \Delta l}{8 D_3^3 n} \quad (22)$$

Where  $\Delta l$  is the amount of elongation of the restraint-spring of the  $j$ th actuator at the bending state. Acquired from Fig. 6,

$$\begin{cases} l_1' = \theta(R_2 - R_3 \cos \beta') \\ l_2' = \theta \left( R_2 - R_3 \cos \left( \frac{\pi}{2} - \beta' \right) \right) \\ l_3' = \theta(R_2 - R_3 \cos(\pi - \beta')) \\ l_4' = \theta \left( R_2 - R_3 \cos \left( \frac{3\pi}{2} - \beta' \right) \right) \\ l_5' = \theta R_2 \end{cases} \quad (23)$$

Through numerical analysis of (23), obtained,

$$l_5' = \frac{l_1' + l_2' + l_3' + l_4'}{4} \quad (24)$$

When the 4 equivalent springs corresponding to four driving cavities were evenly distributed around the actuator,  $1/4$  of the sum of their lengths after deformation was equal to the post-deformation length of the equivalent spring corresponding to the stiffness regulating cavity. If the restraint-spring was evenly divided into  $n_1$  portions along the direction of equivalent spring corresponding to the stiffness adjusting cavity, the number of equivalent springs corresponding to the driving cavity was assumed to be  $n_1$ , which wrapped evenly around the outside of the actuator. The mentioned analysis,  $\Delta l$  could be expressed as,

$$\Delta l = R_2 \theta - L \quad (25)$$

Substituting (2), (4), (5), (21), (22), (25) into (26) and integrate (21), the output force of the  $j$ th actuator at the bending state was obtained.

$$F_j' = \left( \frac{\pi D_2^2}{2} + \frac{\pi D_1^2}{4} \right) P_{jin} - \frac{G_T d_1^4}{8 D_3^3 n} (\theta R_2 - L) - E \left( \frac{\theta R_2}{L} - 1 \right) \frac{(D^2 - D_1^2 - 2 D_2^2)}{4} \pi \quad (26)$$

When the massage head contacted the specified acupoint, the total output force of 3 actuators was equal to the force  $F$  of the human body to the applying force platform. Thus,

$$F' = \sum_{j=1}^3 F_j' \quad (27)$$

#### 4. The workspace of the driving system

The three actuators all generated the moment on the center of the applying force platform in the bending process of the driving system,  $M_1$ ,  $M_2$ , and  $M_3$  respectively. After the moment of the driving system was analyzed,  $\beta$  could be shown in Fig. 7.

$$\beta = \arctan \frac{M_{gy}}{M_{gx}} \quad (28)$$

Where,

$$M_{gx} = (M_2 + M_3) \sin \gamma - M_1, \quad M_{gy} = (M_3 - M_2) \cos \gamma$$

$M_{gx}$  is the component in the X direction of the combined moment,  $M_{gy}$  represents the component in the Y direction of the combined moment,  $\gamma$  expresses the angle of inclusion between the direction of  $M_3$  and the Y-axis. From the geometric relationship in Fig. 7,  $\gamma = \pi/6$ .

The moment generated by the actuator to the applying force platform is,

$$M_j = F_j' \cdot R_1 \quad (29)$$

By combining (17), (26), (28), and (29),

$$\frac{P_{3in} - P_{2in}}{P_{3in} + P_{2in} - 2P_{1in}} = \frac{l_3 - l_2}{l_3 + l_2 - 2l_1} \quad (30)$$

The following law can be obtained by decomposing (30),

$$\begin{cases} k(P_{3in} - P_{2in}) = l_3 - l_2 \\ k(P_{3in} + P_{2in} - 2P_{1in}) = l_3 + l_2 - 2l_1 \end{cases} \quad (31)$$

A linear relationship between  $\Delta l_m$  and  $\Delta P_m$  exists when the driving system is at the bending state according to (31), that is  $\Delta l_k = \Delta P_k \times k$ . Where  $\Delta P_m$  is a pressure difference among the actuators,  $\Delta l_m$  denotes a length difference among the actuators. By combining (9) and (12), the expression for the reciprocal of the scale coefficient could be obtained.

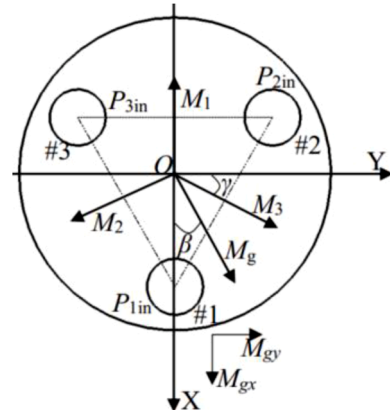


Fig. 7. Bending direction of the driving system.

$$\frac{1}{k} = \frac{E(\pi(D^3 - D_1^3) - 4D_2^3 - 6\pi D_2^2 h_1) + \frac{3\pi L D G_T d_1^4}{2D_3^3 n}}{L(6\pi^2 D D_2^2 + 3\pi^2 D D_1^2 - 4D_2^3 - 6\pi D_2^2 h_1)} \quad (32)$$

$$+ \frac{3\pi^2 D E (D^2 - D_1^2) - 6\pi^2 E D D_2^2}{L(6\pi^2 D D_2^2 + 3\pi^2 D D_1^2 - 4D_2^3 - 6\pi D_2^2 h_1)}$$

The mapping relationship between  $\Delta l_m$  and  $\Delta P_m$  was established by (31). The mapping model between the value of the air pressure input and the bending geometric parameters of the driving system could be built by substituting (31) and (32) into (17), (18). Thus,

$$\beta = \arctan \frac{\sqrt{3}(P_{3in} - P_{2in})}{P_{3in} + P_{2in} - 2P_{1in}} \quad (33)$$

$$\theta = \frac{k\sqrt{2(P_{1in} - P_{2in})^2 + (P_{2in} - P_{3in})^2 + (P_{3in} - P_{1in})^2}}{3R_1} \quad (34)$$

The range of the value of  $\beta$  determined the working range of the driving system, therefore the working range of the driving system could be determined by the value of  $\beta$ . The expression of  $\beta$  under the kinematics model could be written as.

$$\begin{cases} \beta = \arctan f(P_{jin}) \\ f(P_{jin}) = \frac{\sqrt{3}(P_{3in} - P_{2in})}{P_{3in} + P_{2in} - 2P_{1in}} \end{cases} \quad (35)$$

By applying the maximum pressure on actuator #j, the working range of the driving system could be divided into three areas to more effectively calculate the working range of the driving system.

Area I:  $P_{3in}$  is the maximum pressure, that is actuator #3 is the main actuator

Area II:  $P_{2in}$  is the maximum pressure, that is actuator #2 is the main actuator

Area III:  $P_{1in}$  is the maximum pressure, that is actuator #1 is the main actuator

The significance of such a division was that regions I, II, and III represent one workspace of the driving system respectively when one actuator was the primary actuator. The driving system had the same range of bending direction angle in the three zones I, II, and III because of the 3 actuators were evenly distributed between the fixed platform and the applying force platform. Subsequently, the final effect could be achieved by analyzing the working range of anyone. According to Fig. 5, (35) refers to the workspace distribution function built based on region I and then the bending direction angle in regions II and III could be deduced by region I. combining with (35) assumed,

$$\begin{cases} P_{2in} = aP_{3in}, a \in [0, 1] \\ P_{1in} = bP_{3in}, b \in [0, 1] \end{cases} \quad (36)$$

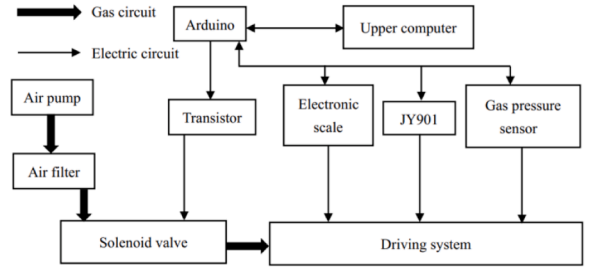
Thus,

$$f(P_{jin}) = \sqrt{3} \frac{1-a}{1+a-2b}, a \in [0, 1], b \in [0, 1] \quad (37)$$

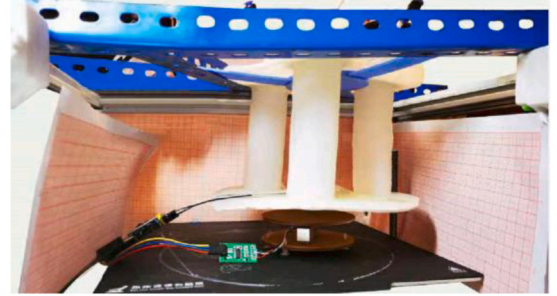
If #3 was defined as the primary actuator and  $P_{2in} = P_{3in}, P_{1in} = 0$ , the projection of the central axis in the X-Y plane was assumed as the position of the 0 degrees bending direction angle.  $\beta \in [0, \frac{2\pi}{3}]$  could be got by solving (35) and (37) in MATLAB. According to the mentioned analysis, the maximum bending direction angle of each region was  $120^\circ$ . Thus, the driving system could deflect in any direction through the coupled motion of three actuators.

## 5. Experiment and analysis

The overall experimental device of the driving system is presented in Fig. 8a, primarily comprising the upper computer, the air pump, the transistor, Arduino, the solenoid valve, the gas pressure sensor, JY901,



(a) Overall experimental device



(b) The experiment at the vertical state



(c) The experiment at the bending state

Fig. 8. Real-time state of the driving system.

and the electronic scale. The power management module provided a 12V DC voltage for the solenoid valve and the air pump, the PWM signal of the Arduino was adopted to control the air pump to output the appropriate air pressure, and the solenoid valve was actuated by controlling the transistor. Gas pressure sensors could collect real-time air pressure numerical within the actuator, JY901 was capable of measuring the bending angle and bending direction angle of the actuator, and the electronic scale could read the overall output force. Fig. 8b, 8c presented the real-time state of the driving system under different air pressures. The model parameters and the values used in the theoretical calculation are listed in Table 1 to compare and analyze the theoretical and experimental values.

In the theoretical calculations, Young's modulus of silica gel more

Table 1  
Model parameters of the driving system.

|           |          |           |
|-----------|----------|-----------|
| Parameter | $L$      | $D$       |
| Value     | 0.12 m   | 0.05 m    |
| Parameter | $D_1$    | $D_2$     |
| Value     | 0.02 m   | 0.02 m    |
| Parameter | $D_3$    | $n$       |
| Value     | 0.05 m   | 60        |
| Parameter | $d_1$    | $h_1$     |
| Value     | 0.0006 m | 0.012 m   |
| Parameter | $R_1$    | $G_T$     |
| Value     | 0.065 m  | 77690 Mpa |

significantly impacted the calculation results. The theoretical formula of  $E$  could not be directly yielded due to the material properties of silica gel affected by the processing technology. The tensile tests could be performed on a high-low temperature double-column tester to determine  $E$ . The ratio of tensile strength and elongation of 20-degree silica gel was examined many times, lastly, the value of  $E$  was set at 0.7 Mpa.

If the driving system was in a vertical state, the reading of the electronic scale would be cleared to zero when the driving system contacted the electronic scale surface and maintained balance to eliminate the effect of gravity on the experimental results, as shown in Fig. 8b. Fig. 8c presents the bending state of the driving system. If the output force of the driving system was measured at this state, the angle of deflection of the electronic scale would be manually regulated and the reading of the electronic scale would be cleared to zero.

### 5.1. Predictive models for workspaces

In the massage process, the working space acted as an important index to measure the performance of the massage robots. The three actuators should cooperate with each other to complete different massage movements. If the massage robot performed the palm pressing operation, the driving system could do this by simply inflating and deflating the air. Thus, this paper primarily analyzed the working space of the driving system when the massage robot performed the palm kneading operation. In the theoretical calculation, by keeping the air pressure of one of the actuators constant and regulating the air pressure of the other 2 actuators, the driving system could be rotated in all directions and  $\theta$  could reach  $180^\circ$ . By complying with the theoretical model of the driving system workspace to conduct the numerical

calculation, the maximum workspace could be obtained theoretically. As shown in Fig. 9.

The circle arcs in Fig. 9a represents the fitting curve of the position of the central axis vertex with  $\theta$  reaching  $180^\circ$  and the space formed by three circle arcs is the overall working space of the driving system theoretically. The outermost circle is formed by the furthest point of the central axis vertex of the driving system theoretically. The space composed of polka dots in Fig. 9b represents the overall workspace of the driving system in the  $0-180^\circ$  bending process. Fig. 9a illustrates the maximum theoretical workspace of the driving system in the 3D view, and Fig. 9b presents the projection of the workspace on the X-Y plane. The working range of the driving system could be understood better from different perspective. The intersection coordinates of the three fitting circle arcs and the coordinates of the farthest point that the vertex of the center axis could reach were listed in Table 2.

### 5.2. Accuracy verification of the predictive model of the workspace

A bending direction angle was selected for the experiment in regions I, II, and III, respectively. Subsequently, the theoretical results and experimental results were analyzed to verify the accuracy of the prediction model of the workspace. When the experimental data of the bending angle was being collected, the driving system was run to the position of the specified bending direction angle firstly. Then, the air pressure in the three actuators was debugged constantly to make the bending deformation occur, but the bending direction angle unchanged. The bending angles of the 3 specified positions were set in the range of  $0-30^\circ$ , the data of the vertex position of the central axis was collected once every bending  $3^\circ$ , and the experiment was repeated three times. The maximum bending angle of the driving system was  $67^\circ$ . The bending angle was controlled to be tested in the range of 0 to  $30^\circ$ , with the main purpose of verifying the accuracy of the prediction model for the working interval. When the experimental data were being processed, the first point, the midpoint, and the endpoint were selected as the designated points to determine the error between the numerical results and the experimental results. The error value was the Euclidean distance between the mean value of the three experimental results and the theoretical results, as shown in Fig. 10.

Fig. 10a, 10b draws the comparison results between the experimental and theoretical values of the driving system's workspace. It could be seen that the experimental points were distributed around the theoretical curve, which demonstrated that the driving system deformed steadily in one direction at the fixed bending direction angle. The obvious deviation did not occur in the deformation, thereby indicating the good stability of the driving system. Fig. 10c presents the error between the theoretical and experimental values of the driving system and when the bending direction angle was fixed, and in the process of the bending angle reached  $30^\circ$ , the maximum error was 10.764 mm, and the minimum error was 2.78 mm. Moreover, the midpoint error was the maximum error. The error was explained that the silica gel exhibited the viscosity, the amount of elongation of the actuator at the beginning of the inflation was smaller than the theoretical value, and when the air pressure reached a threshold value, the sensitivity of silica gel to air pressure would tend to decrease.

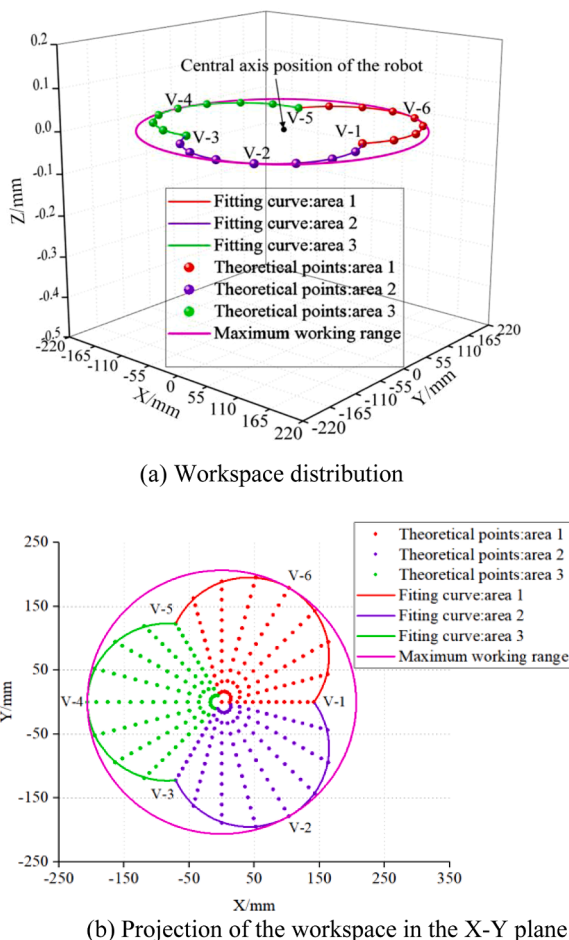
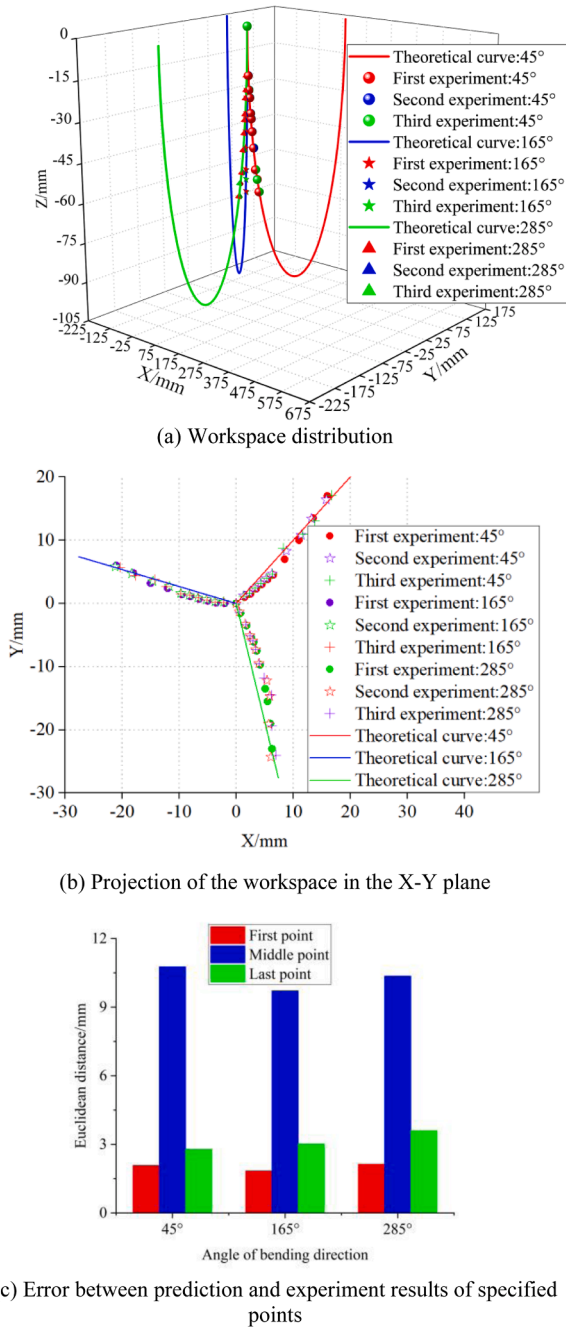


Fig. 9. Workspace of the central axis of the driving system.

Table 2  
Coordinates of six special point.

|      | V-1     | V-2     | V-3     |
|------|---------|---------|---------|
| X/mm | 141.4   | 103.21  | -70.7   |
| Y/mm | 0       | -178.74 | -122.46 |
| Z/mm | 0       | 0       | 0       |
|      | V-4     | V-5     | V-6     |
| X/mm | -206.38 | -70.7   | 103.21  |
| Y/mm | 0       | 122.46  | 178.74  |
| Z/mm | 0       | 0       | 0       |



**Fig. 10.** Comparison between prediction and experiment results of the workspace.

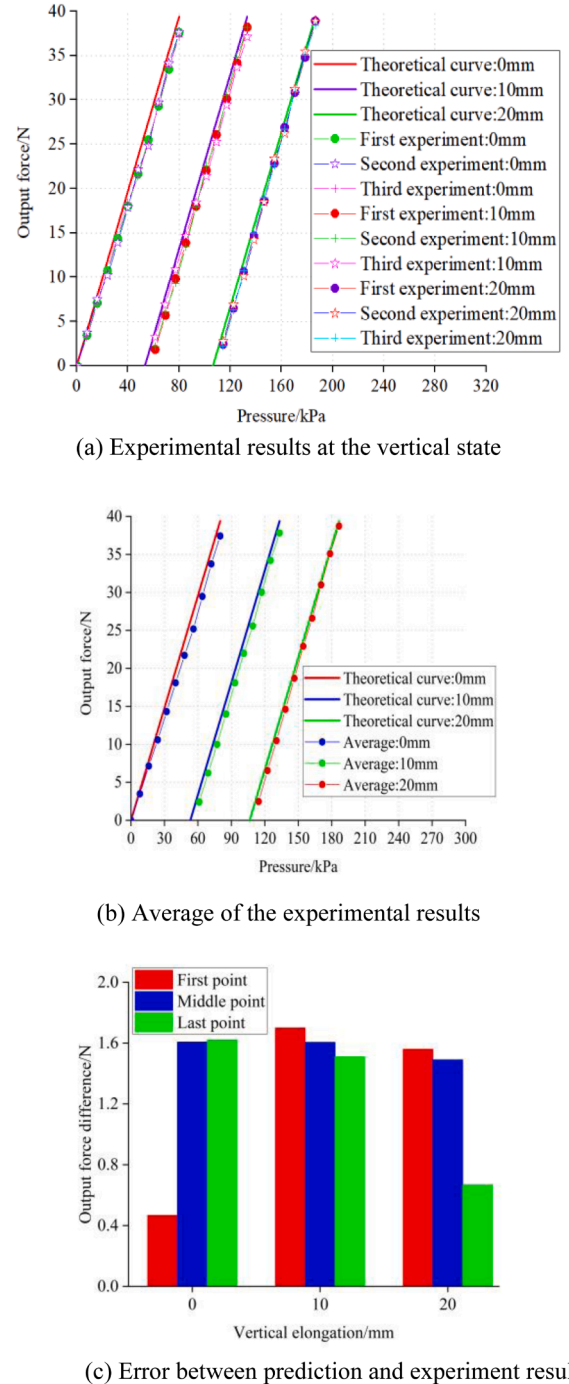
### 5.3. Output force of the driving system

The output force of the soft massage robot was a vital standard to measure whether it could meet people's expectations. The real-time adjustment of the output force could be critical to the process of massage. This paper mainly verified the output force model at the vertical state and the bending state. When verifying the output force at the vertical state, the driving system contacted the surface of the electronic scale and there was no mutual force between the two, then the same air pressure was charged into the three actuators. When verifying the output force at the bending state, it was ensured that the driving system contacted the surface of the electronic scale and there was no mutual force between the two. Lastly, the air pressure value of the three actuators was constantly regulated.

#### 5.3.1. The output force at the vertical state

When the driving system was at the vertical state and the initial amount of elongation was 0 mm, 10 mm, or 30 mm respectively, the three actuators were pressurized and the single-step pressurization reached 8 kPa. Subsequently, the output force at different air pressure could be determined. When the experimental data were being processed, the output forces corresponding to air pressure increments of 8 kPa, 40 kPa, and 80 kPa were used as the reference values to determine the error between the theoretical value and the experimental value. The error value was the absolute value of the difference between the mean value of the 3 experiments and the theoretical value, as shown in Fig. 11.

Fig. 11a, 11b presents the output force characteristics of the driving



**Fig. 11.** Comparison between prediction and experiment results of output force.



system at the vertical state that a relatively large output force could be provided. According to Fig. 11a, 11b, the experimental values were all distributed below the theoretical curve, and with the increase in the air pressure, the absolute value of the difference between the experimental values and the theoretical curve increased firstly, and then it tended to be smooth and finally decreased gradually. As indicated from Fig. 11a, 11c, the error of the output force increased firstly, then turned approximately equal, and finally decreased, which was consistent with the mentioned law. According to Fig. 11b, the maximum error occurred when the driving system with an initial length of 0mm was pressurized to 56kPa, and the error value was 2.2N. The fitted experimental curve showed a marked tendency to decrease when the driving system with an initial length of 10 mm was pressurized to 109kPa, and the error value was 1.8N.

The reason for the mentioned phenomenon was that the silica gel exhibited the viscosity. The amount of elongation of the actuator at the beginning of the inflation was smaller than the theoretical value and as the air pressure increased to 56kPa, the speed of elongation of silica gel increased, and then the amount of elongation began to slowly approach the normal value. However, the sensitivity of silica gel to air pressure would gradually decrease and the amount of the elongation of the actuator was less than the theoretical value again with the air pressure reaches a threshold value (109 kPa). In addition, the restraint-spring wrapping around the outer wall elongated in the elongation of the actuator, thereby causing some friction resistance to the elongation of the actuator. Although the actuator was wrapped with a restraint-spring on the outside, a certain degree of radial deformation was still generated at the pitch of the restraint-spring. The two mentioned factors were ignored in the theoretical modeling. Thus, this paper proposed the concept of "correction function". Subtracting a "correction function" from the theoretical model to make the theoretical value and experimental value equal to the greatest extent.

① Initial amount of elongation: 0 mm

$$F_c = \begin{cases} F - 0.0957P_{in}^{0.7748} & (0 \text{ kPa} \leq P_{in} \leq 40 \text{ kPa}) \\ F - 0.5855e^{0.025P_{in}} & (40 \text{ kPa} \leq P_{in} \leq 56 \text{ kPa}) \\ F - 4.7772e^{-0.0128P_{in}} & (56 \text{ kPa} \leq P_{in} \leq 80 \text{ kPa}) \end{cases} \quad (38)$$

② Initial amount of elongation: 10 mm

$$F_c = \begin{cases} F - 0.0002P_{in}^2 + 0.0452P_{in} - 30.3565 & (61.25 \text{ kPa} \leq P_{in} \leq 109.25 \text{ kPa}) \\ F + 0.016P_{in} - 29.5828 & (109.25 \text{ kPa} \leq P_{in} \leq 133.25 \text{ kPa}) \end{cases} \quad (39)$$

③ Initial amount of elongation: 20 mm

$$F_c = \begin{cases} F + 0.0179P_{in} - 56.0665 & (114.5 \text{ kPa} \leq P_{in} \leq 146.5 \text{ kPa}) \\ F - \sqrt{46905.93 - P_{in}^2} + 299.46P_{in} + 209.7414 & (146.5 \text{ kPa} \leq P_{in} \leq 162.5 \text{ kPa}) \\ F + 0.0161P_{in} - 55.8018 & (162.5 \text{ kPa} \leq P_{in} \leq 186.5 \text{ kPa}) \end{cases} \quad (40)$$

Where  $P_{in}$  is the air pressure value (kPa) existing in the actuator,  $F_c$  represents the function expression modified. According to Fig. 12, the experimental values were almost identical to the correction values.

### 5.3.2. Output force in bending state

The bending state of the driving system could be divided into two states: ① the bending angle was unfixed; ② the bending angle was fixed. When the first case was investigated, only the output force was explored,

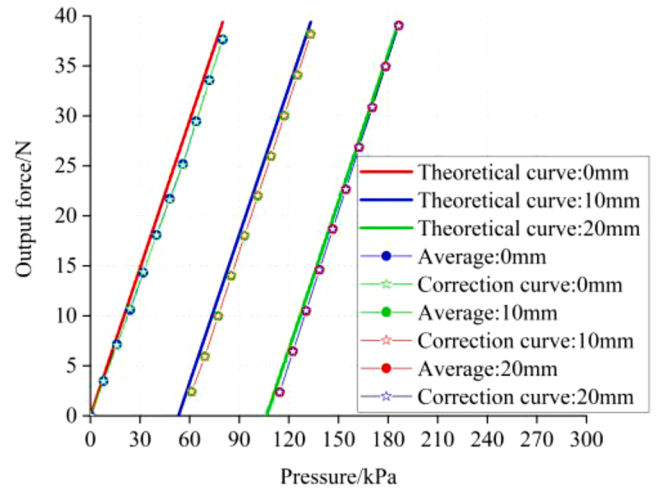
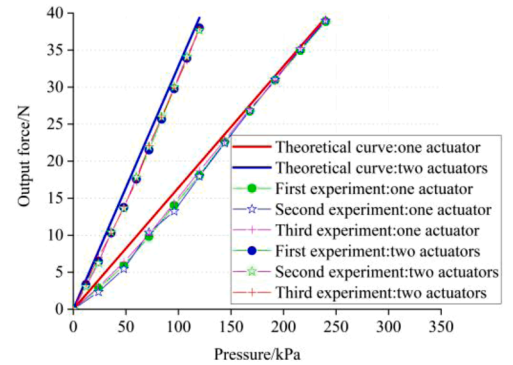
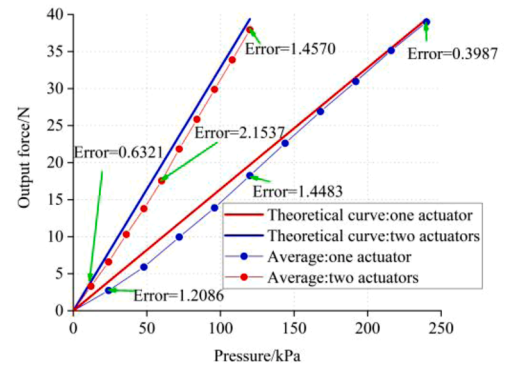


Fig. 12. Modify values at the vertical state.

the bending angle and the bending direction angle were not considered. When the second scenario was researched, the bending direction angle would also be fixed. In this paper, the case of inflating a single actuator and filling two actuators with an equal amount of gas was studied to verify the error of the first case. The error of the bending angle at the fixed state could be analyzed by fixing the bending angle at  $15^\circ$  and  $25^\circ$ . When the experimental data were being processed, the first data, the intermediate data, and the last data of the measured value were all taken as the reference value to determine the error between numerical results and experimental results. The error value was the absolute value of the difference between the mean value of the three experimental results and



(a) Experimental results of bending angle at an unfixed state



(b) Average of the experimental results

Fig. 13. Comparison between theoretical and experiment value of output force.

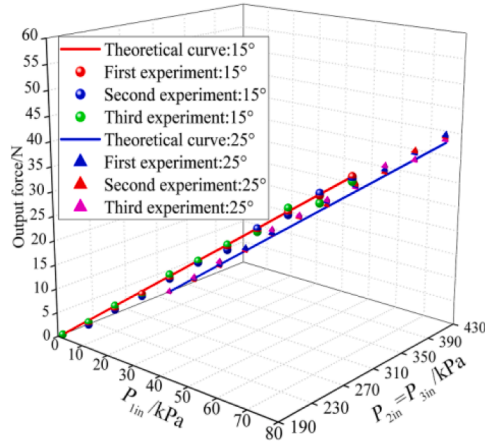
the theoretical value, as shown in Figs. 13, 14.

In this paper, when studying the output force of the driving system with a fixed bending angle, the following rules were formulated.

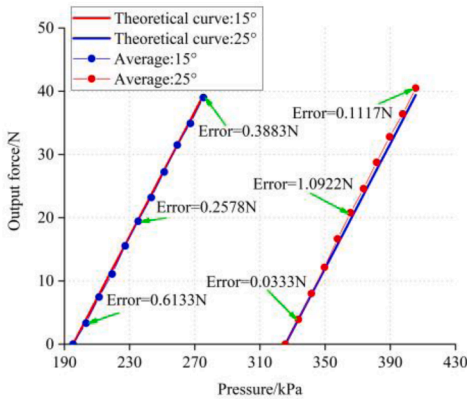
- I If the driving system is tilted toward the region I, set the air pressure in actuator #1 at 0kPa and actuator #2, #3 are charged into the same air pressure.
- II If the driving system is tilted toward the region II, set the air pressure in actuator #2 at 0kPa and actuator #1, #3 are charged into the same air pressure.
- III If the driving system is tilted toward the region III, set the air pressure in actuator #3 at 0kPa and actuator #1, #2 are charged into the same air pressure.

Area 1 was chosen as the research object in this paper, at the bending angle of 15°, the initial air pressure of the three actuators was,  $P_{1in}=0$  kPa,  $P_{2in}=P_{3in}=195.3$  kPa. At the bending angle of 25°, the initial air pressure of the three actuators was,  $P_{1in}=0$  kPa,  $P_{2in}=P_{3in}=325.6$  kPa.

Fig. 13 shows the output force characteristics in the first case, and Fig. 14 presents the output force characteristics in the second scenario, thereby demonstrating that a relatively large output force could be provided at both states. Observing Fig. 13, the variation law of output force was identical to Fig. 11. As indicated in Fig. 14, when the air pressure ranged from 195 to 350 kPa, the experimental and theoretical values of output forces were roughly the same. The average error was 0.4197N, which was relative to the average output force (21.0943N) less than 2% at this stage. As the air pressure increased further, the error



(a) Experimental results of bending angle at a fixed state  
b Average of the experimental results



(b) Average of the experimental results

Fig. 14. Comparison between theoretical and experiment value of output force.

between the theoretical and experimental values of the output force of the driving system increased gradually and the actual output force also began to exceed the theoretical output force. The reason for this was that the actual wall thickness of the actuator was different at the respective position during the bending process, so the Young's modulus of the silica gel would change. When the air pressure reached a threshold value (109 kPa), the sensitivity of silica gel to air pressure would gradually decrease and the amount of the elongation of the actuator was lower than the theoretical value. Meanwhile, with the air pressure increasing in the stiffness adjusting cavity, the load-bearing bending moment of the actuator became larger and the ability to resist the external factors affecting the deformation of the actuator was stronger. In the actual experiment, when the actuator was bent and deformed, the restraint-spring wrapped on the outer wall also bent, thereby causing some frictional resistance to the bending of the actuator. At the same time, a certain degree of radial deformation would occur when the actuator was being bent. The two factors mentioned above were ignored in the theoretical modeling.

Given the mentioned analysis, a "correction function" was presented for the output force in the range of 0–195kPa and 350–405.6 kPa. Meanwhile, the output force at the bending state was rewritten as,

$$F'_j = f(P_{jin}) \quad (41)$$

#### ① Inflate a single actuator

$$F_C = \begin{cases} \sum_{j=1}^3 F'_j - 0.0413P_{in} - 0.0721 (0 \text{ kPa} \leq P_{in} \leq 48 \text{ kPa}) \\ \sum_{j=1}^3 F'_j + 0.0062P_{in} - 2.3014 (48 \text{ kPa} \leq P_{in} \leq 195 \text{ kPa}) \end{cases} \quad (42)$$

Where,  $P_{3in}=P_{in}$ ,  $P_{1in}=P_{2in}=0$ .

#### ② Inflate the two actuators

$$F_C = \begin{cases} \sum_{j=1}^3 F'_j - 0.0001P_{in}^2 - 0.041P_{in} - 0.0309 (0 \text{ kPa} \leq P_{in} \leq 36 \text{ kPa}) \\ \sum_{j=1}^3 F'_j - 2.4308e^{-0.9332e^{-0.0253P_{in}}} (36 \text{ kPa} \leq P_{in} \leq 72 \text{ kPa}) \\ \sum_{j=1}^3 F'_j + 0.0141P_{in} - 3.0807 (72 \text{ kPa} \leq P_{in} \leq 120 \text{ kPa}) \end{cases} \quad (43)$$

Where,  $P_{3in}=P_{2in}=P_{in}$ ,  $P_{1in}=0$ .

#### ③ Inflate the three actuators

$$F_C = \begin{cases} \sum_{j=1}^3 F'_j - 0.0004P_{in}^2 + 0.3264P_{in} - 171.8487 \\ (350 \text{ kPa} \leq P_{in} \leq 389.6 \text{ kPa}) \\ \sum_{j=1}^3 F'_j - 0.006P_{in} - 103.5146 (389.6 \text{ kPa} \leq P_{in} \leq 405.6 \text{ kPa}) \end{cases} \quad (44)$$

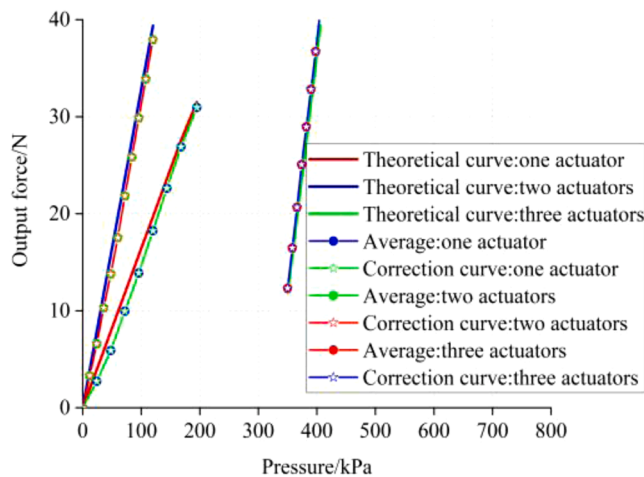


Fig. 15. Modify values in bending state.

Where,  $P_{1in}=P_{in}-325.6$ ,  $P_{2in}=P_{3in}=P_{in}$ . According to Fig. 15, the experimental values were almost identical to the correction values.

## 6. Conclusion

In this paper, one driving system of the soft massage robot was presented, consisting of three actuators, a fixed platform, and an applying force platform. The driving system was driven by air pressure and it could switch between curved and vertical states, thereby completing palm kneading and palm pressing. On that basis, the kinematics model, the force model, and the workspace model of the driving system were built, and a set of test equipment was set up to verify the accuracy of the theoretical calculation.

In the experiment, the accuracy of the prediction model was verified, and the stability of the pneumatic module was examined by performing the repeated test. The maximum and minimum output forces at the vertical state at the range of 0–186.5kPa were 38.72N and 37.42N, respectively. In the pressure range of 0–240kPa, when the bending angle was not fixed, the maximum and minimum output forces of the driving system reached 38.98N and 37.93N, respectively, while the values were 40.49N and 38.99N, respectively, under 0–405.6kPa and at the fixed bending angle. All of the mentioned cases could satisfy the output force required by the massage. The driving system proposed in this paper was not only simple in structure but also could provide the output force required by the massage robot. Besides, its structure size could be set in accordance with the actual situation. Under the control of the driving system, the soft massage robot proposed in this paper could complete two actions: the palm kneading and the palm pressing. Since aching backs have become a common disease, the soft massage robot studied in this paper could satisfy the requirements of easing back pain, the subsequent development is worth prospecting.

## CRediT authorship contribution statement

**Yong Liu:** Conceptualization, Methodology, Software, Validation, Writing – review & editing. **Xiaojing Yang:** Resources, Supervision, Funding acquisition. **RuiYing Su:** Validation, Formal analysis, Investigation, Data curation, Writing – original draft.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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**Update**

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Corrigendum

Corrigendum to ‘Design and modeling of a driving system for soft massage robot’ [79 (2021) 102640]<sup>☆</sup>

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The authors regret that they miswrote the "minus sign" of the equation (12) as the "plus sign".

Wrong symbol in the original article:

$$F_j = \frac{Ed}{\pi DL} \left( \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2} \right) + \left( \frac{D_2^3}{3} + \frac{\pi D_2^2 h_1}{2} \right) \frac{P_{jin}}{\pi D}$$

Correct equation should be:

$$F_j = \frac{Ed}{\pi DL} \left( \frac{(D^3 - D_1^3)}{12} \pi - \frac{D_2^3}{3} - \frac{\pi D_2^2 h_1}{2} \right) - \left( \frac{D_2^3}{3} + \frac{\pi D_2^2 h_1}{2} \right) \frac{P_{jin}}{\pi D}$$

The authors would like to apologise for any inconvenience caused.  
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